

Lab – 10

Some Advance Features

November 14, 2024

Transactions

Transactions (1)

- A sequential group of database manipulation operations, performed as a one single work unit.
- A transaction will never be complete unless each individual operation within the group is successful.
- If any operation within the transaction fails, the entire transaction will fail.
- Example: Banking transaction, a transfer of 100.
 - Check that the balance of 1st account is > 100 .
 - Deduct 100 from the 1st account.
 - Add 100 to the 2nd account.

Transactions (2)

- In SQL, we have a session variable AUTOCOMMIT.
- If AUTOCOMMIT is set to 1 (default in MySQL):
 - Each SQL statement is seen as a complete transaction.
 - Each transaction is automatically made permanent.
- If AUTOCOMMIT is set to 0:
 - Subsequent series of statements acts like a transaction.
 - Activities are committed by an explicit COMMIT statement.
- In MySQL, transactions begin with the statement BEGIN and end with a COMMIT or ROLLBACK statement.

Transactions (3)

AUTOCOMMIT = 1;

BEGIN; —————→ AUTOCOMMIT = 0;

semicolon separated bulk of SQL commands

COMMIT OR ROLLBACK; —→ AUTOCOMMIT = 1;

Properties of Transactions

- **ACID** - 4 standard properties are:
 - **Atomicity** - All operations are completed successfully.
 - **Consistency** - Database properly changes the states.
 - **Isolation** - Transactions should operate independently and transparently to each other.
 - **Durability** - The result of a committed transaction persists.
- For successful transaction issue COMMIT command.
- If a failure occurs, a ROLLBACK command should be issued.

CONNECTION 1 : Green

CONNECTION 2 : Blue

BEGIN;

INSERT INTO t (f) VALUES (1);

SELECT * FROM t;

An Empty Set

COMMIT;

SELECT * FROM t;

Updated Set

CONNECTION 1 : Green

CONNECTION 2 : Blue

BEGIN;

INSERT INTO t (f) VALUES (1);

SELECT * FROM t;

An Empty Set

ROLLBACK;

SELECT * FROM t;

Empty Set

Another Transaction Example

- Check the account balance for customer and terminate the account if its balance is less than 500. If the account is older than year 2010 the transaction will “COMMIT” else transaction will “ROLLBACK”

Revisit Your Database For Answer

Depositor Table

Customer_name	Account_number
Priya	102
Yash	101
Yash	201
Vinay	217
Anjali	222
Divya	217
Rohit	305

Account Table

Account_Number	Branch_Name	Balance	Date
101	Wright Town	500	5-2-11
215	Mehgawan	700	3-7-12
102	S street	400	6-8-10
305	Napier town	350	4-6-09
201	Stadium	900	9-4-10
222	Cross square	700	8-11-11
217	Stadium	750	2-10-12

Another Transaction Example – Contn..

- The code will check the account balance for customer and terminate the account if its balance is less than 500. If the account is older than year 2010 the transaction will “COMMIT” else transaction will “ROLLBACK”
- Run **front.php**
- See Code for Transaction in **transaction.php**
- Run code for customers :
 - Priya - **ROLLBACK**
 - Rohit - **COMMIT**

Accessing Database with Various Privilege Levels

Assigning Privileges

Granting access to users on your database:

- You should have appropriate privileges to assign rights on your own database.
- A trial database has been created with super-user's privilege.
- Consider following 5 users with widening scope of their rights on 'trial' database:

USER	Privilege
user1	NO Privilege
user2	SELECT
user3	SELECT, INSERT
user4	SELECT, INSERT, UPDATE
user5	SELECT, INSERT, UPDATE, DELETE

Assigning Privileges by Queries (1)

- The syntax to create any user in MySQL –
 - CREATE USER *user_specification* [, *user_specification*] ...
- user_specification: user [IDENTIFIED BY [PASSWORD] 'password']
- Ex – Suppose we want to create 'user1' with password 'password' -

CREATE USER 'user1'@'localhost' IDENTIFIED BY 'password';

Note - To create user who can connect remotely to MySQL use '%' instead.

Assigning Privileges by Queries (2)

- The syntax to GRANT access to user in MySQL –
 - GRANT privilege_type ON
{ table_Name | Database_Name.* } TO USER
- Ex – Suppose we want 'user3' to give only SELECT,INSERT access to all the tables in database 'trial'

```
GRANT SELECT,INSERT ON trial.* TO  
'user3'@'localhost';
```

Assigning Privileges by Queries (3)

- Create users

```
CREATE USER 'user1'@'localhost' IDENTIFIED BY 'pass1';
```

```
CREATE USER 'user2'@'localhost' IDENTIFIED BY 'pass2';
```

```
CREATE USER 'user3'@'localhost' IDENTIFIED BY 'pass3';
```

```
CREATE USER 'user4'@'localhost' IDENTIFIED BY 'pass4';
```

```
CREATE USER 'user5'@'localhost' IDENTIFIED BY 'pass5';
```

- Grant privilege

```
// NO PRIVILAGE GRANT ON trial.* TO 'user1'@'localhost';
```

```
GRANT SELECT ON trial.* TO 'user2'@'localhost';
```

```
GRANT SELECT,INSERT ON trial.* TO 'user3'@'localhost';
```

```
GRANT SELECT,INSERT,UPDATE ON trial.* TO 'user4'@'localhost';
```

```
GRANT SELECT,INSERT,UPDATE,DELETE ON trial.* TO 'user5'@'localhost';
```


Assigning Privileges through Interface

Click on Databases

The screenshot shows the phpMyAdmin web interface in a browser window. The browser's address bar displays the URL `localhost/phpmyadmin/index.php?route=/&lang=en`. The phpMyAdmin interface has a top navigation bar with several tabs: **Databases** (highlighted with a blue circle), **SQL**, **Status**, **User accounts**, **Export**, **Import**, **Settings**, **Replication**, **Variables**, **Charsets**, and **More**. On the left side, there is a sidebar with the phpMyAdmin logo, a 'Recent' and 'Favorites' section, and a tree view of databases including 'bank_db', 'information_schema', 'mysql', 'performance_schema', 'phpmyadmin', and 'test'. The main content area is divided into several panels: 'General settings' (showing 'Server connection collation' as 'utf8mb4_unicode_ci'), 'Appearance settings' (showing 'Language' as 'English' and 'Theme' as 'pmahomme'), 'Database server' (listing server details like 'Server: 127.0.0.1 via TCP/IP', 'Server type: MariaDB', 'Server version: 10.4.24-MariaDB', etc.), 'Web server' (listing 'Apache/2.4.53 (Win64) OpenSSL/1.1.1n PHP/8.1.6'), and 'phpMyAdmin' (showing 'Version information: 5.2.0' and a link to 'Documentation'). The Windows taskbar at the bottom shows the search bar, several application icons, and the system clock indicating 14:43 on 08-11-2023.

Select a Database and Check Privileges

The screenshot shows the phpMyAdmin web interface in a browser. The address bar indicates the URL is `localhost/phpmyadmin/index.php?route=/server/databases`. The interface has a top navigation bar with tabs for Databases, SQL, Status, User accounts, Export, Import, Settings, Replication, Variables, Charsets, and More. Below this is a 'Databases' section with a 'Create database' form and a table of existing databases. The left sidebar shows a tree view of databases, with 'performance_schema' selected. The main table lists databases with their collations and a 'Check privileges' link for each. The 'Check privileges' link for 'performance_schema' is circled in blue.

Server: 127.0.0.1

Databases | SQL | Status | User accounts | Export | Import | Settings | Replication | Variables | Charsets | More

Databases

Create database

Database name: utf8mb4_general_ci Create

☐ Check all Search

	Database	Collation	Action
☐	bank_db	utf8mb4_general_ci	Check privileges
☐	information_schema	utf8_general_ci	Check privileges
☐	mysql	utf8mb4_general_ci	Check privileges
☐	performance_schema	utf8_general_ci	Check privileges
☐	phpmyadmin	utf8_bin	Check privileges
☐	test	latin1_swedish_ci	Check privileges

Console

Type here to search

31°C Haze 14:44 08-11-2023

Click on Edit Privileges of a Particular User to Edit it

The screenshot shows the phpMyAdmin web interface in a browser. The address bar indicates the URL: `localhost/phpmyadmin/index.php?route=/server/privileges&db=bank_db&checkprivsdb=bank_db`. The interface displays the 'Privileges' tab for the 'bank_db' database. A table lists users with access to the database. The 'Edit privileges' link for the 'root' user is circled in blue.

Server: 127.0.0.1 » Database: bank_db

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers More

Users having access to "bank_db"

	User name	Host name	Type	Privileges	Grant	Action
☐	root	127.0.0.1	global	ALL PRIVILEGES	Yes	Edit privileges Export
☐	root	::1	global	ALL PRIVILEGES	Yes	Edit privileges Export
☐	root	localhost	global	ALL PRIVILEGES	Yes	Edit privileges Export

☐ Check all With selected: [Export](#)

New

[Add user account](#)

Console

You can Select/Deselect Privileges according to Your Preference

The screenshot shows the phpMyAdmin interface in a web browser. The address bar indicates the URL: `localhost/phpmyadmin/index.php?route=/server/privileges&username=root&hostname=127.0.0.1&dbname=&tablename=&routinename=`. The browser's taskbar at the bottom shows various open applications including Gmail, YouTube, Maps, New chat, SCISPACE, deepfake detection..., and GitHub - Daisy-Zha....

The phpMyAdmin interface has a sidebar on the left with a tree view of databases and schemas. The main content area is titled "Edit privileges: User account 'root'@'127.0.0.1'". It features a tabbed interface with tabs for Databases, SQL, Status, User accounts, Export, Import, Settings, Replication, Variables, Charsets, and More. The "User accounts" tab is active.

Under the "User accounts" tab, there is a section for "Global privileges" with a "Check all" button. Below this, a note states: "Note: MySQL privilege names are expressed in English." There are four main categories of privileges, each with a list of specific privileges and checkboxes:

- Data** (checked):
 - ☒ SELECT
 - ☒ INSERT
 - ☒ UPDATE
 - ☒ DELETE
 - ☒ FILE
- Structure** (checked):
 - ☒ CREATE
 - ☒ ALTER
 - ☒ INDEX
 - ☒ DROP
 - ☒ CREATE TEMPORARY TABLES
 - ☒ SHOW VIEW
 - ☒ CREATE ROUTINE
 - ☒ ALTER ROUTINE
 - ☒ EXECUTE
 - ☒ CREATE VIEW
 - ☒ EVENT
 - ☒ TRIGGER
- Administration** (checked):
 - ☒ GRANT
 - ☒ SUPER
 - ☒ PROCESS
 - ☒ RELOAD
 - ☒ SHUTDOWN
 - ☒ SHOW DATABASES
 - ☒ LOCK TABLES
 - ☒ REFERENCES
 - ☒ REPLICATION CLIENT
 - ☒ REPLICATION SLAVE
 - ☒ CREATE USER
- Resource limits**:
 - Note: Setting these options to 0 (zero) removes the limit.
 - MAX QUERIES PER HOUR: 0
 - MAX UPDATES PER HOUR: 0
 - MAX CONNECTIONS PER HOUR: 0
 - MAX USER_CONNECTIONS: 0

At the bottom of the interface, there is a "Console" tab and a Windows taskbar showing the system clock as 14:44 on 08-11-2023, with a temperature of 31°C and weather conditions of Haze.

Click on Add User Account to Create a New User

The screenshot shows the phpMyAdmin web interface. The browser address bar indicates the URL: `localhost/phpmyadmin/index.php?route=/server/privileges&db=bank_db&checkprivsdb=bank_db`. The interface is for the 'bank_db' database. The 'Privileges' tab is selected, displaying a table of users with access to the database. Below the table, the 'Add user account' button is circled in blue.

Server: 127.0.0.1 » Database: bank_db

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers More

Users having access to "bank_db"

	User name	Host name	Type	Privileges	Grant	Action
☐	root	127.0.0.1	global	ALL PRIVILEGES	Yes	Edit privileges Export
☐	root	::1	global	ALL PRIVILEGES	Yes	Edit privileges Export
☐	root	localhost	global	ALL PRIVILEGES	Yes	Edit privileges Export

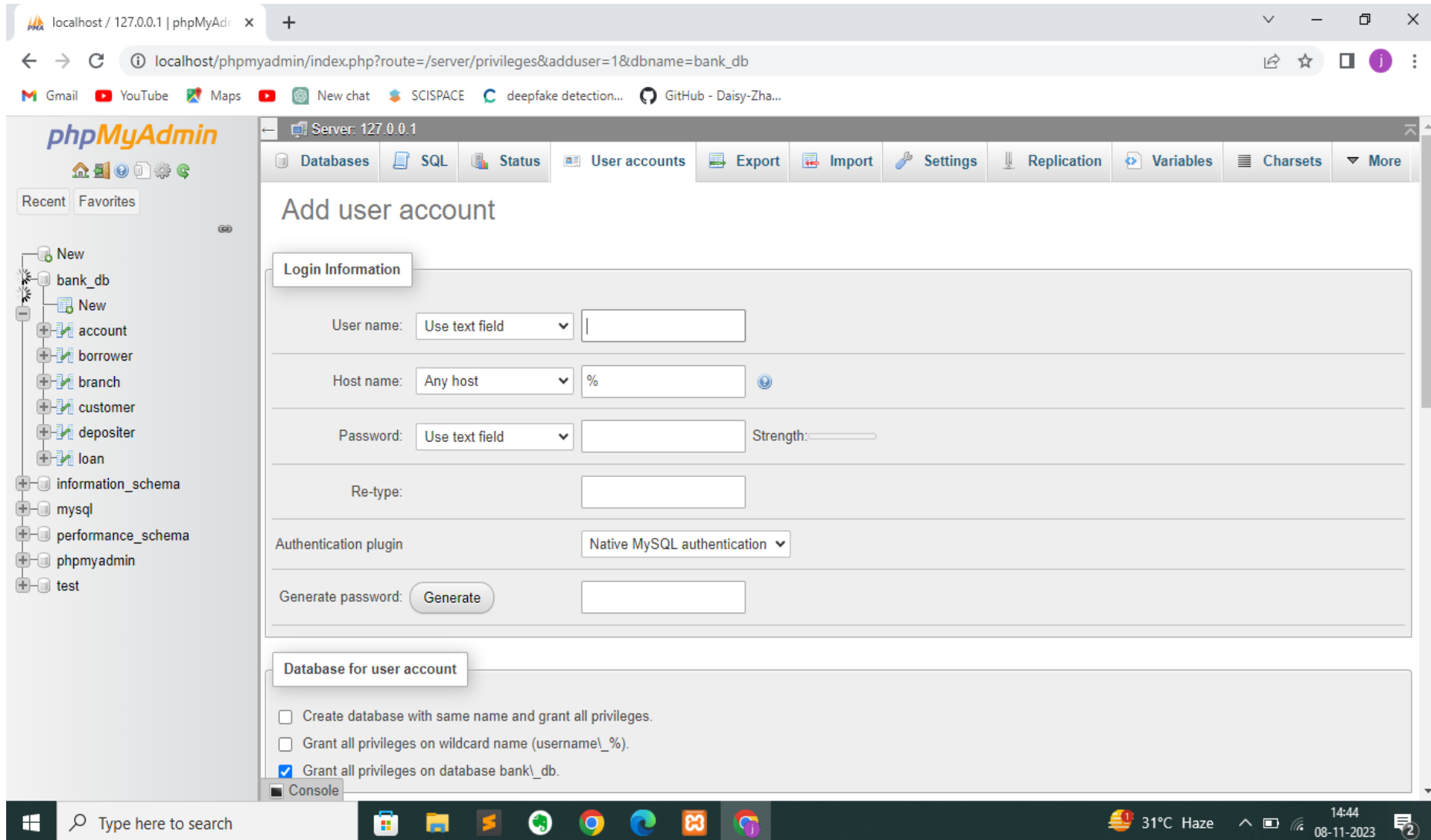
☐ Check all With selected: [Export](#)

New

[Add user account](#)

Console

Set Privileges for the New User



The screenshot shows the phpMyAdmin interface in a web browser. The address bar indicates the URL: `localhost/phpmyadmin/index.php?route=/server/privileges&adduser=1&dbname=bank_db`. The browser's taskbar at the bottom shows various open applications and the system clock.

The phpMyAdmin interface has a sidebar on the left with a tree view of databases. The main content area is titled "Add user account" and is divided into two sections:

- Login Information**: This section contains several input fields and a dropdown menu.
 - User name**: A dropdown menu with "Use text field" selected and an adjacent text input box.
 - Host name**: A dropdown menu with "Any host" selected and an adjacent text input box containing "%".
 - Password**: A dropdown menu with "Use text field" selected and an adjacent text input box. To the right of the password field is a "Strength" indicator.
 - Re-type**: A text input box for confirming the password.
 - Authentication plugin**: A dropdown menu with "Native MySQL authentication" selected.
 - Generate password**: A button labeled "Generate" next to a text input box.
- Database for user account**: This section contains three checkboxes for granting privileges.
 - ☐ Create database with same name and grant all privileges.
 - ☐ Grant all privileges on wildcard name (username_%).
 - ☒ Grant all privileges on database bank_db.

At the bottom of the main content area, there is a "Console" tab.

For Table Privileges

The screenshot shows the phpMyAdmin web interface in a browser. The address bar indicates the URL: `localhost/phpmyadmin/index.php?route=/server/privileges&db=bank_db&table=account&checkprivsdb=bank_db&checkprivstable=account&viewing_mode=table`. The breadcrumb navigation at the top reads: `Server: 127.0.0.1 » Database: bank_db » Table: account`. The left sidebar shows the database structure with `bank_db` expanded, listing tables like `account`, `borrower`, `branch`, `customer`, `depositer`, and `loan`. The main panel displays the 'Privileges' tab for the `account` table. A title bar above the table reads 'Users having access to "bank_db.account"'. The table lists three users with 'root' as the username, all with 'global' type and 'ALL PRIVILEGES'. Each row includes 'Edit privileges' and 'Export' links. Below the table, there is a 'Check all' checkbox and an 'Export' link for the selected items. A 'New' button is visible below the table, with a link to 'Add user account'. The bottom of the image shows the Windows taskbar with the search bar and system tray.

phpMyAdmin

Recent Favorites

Server: 127.0.0.1 » Database: bank_db » Table: account

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers

Users having access to "bank_db.account"

	User name	Host name	Type	Privileges	Grant	Action
☐	root	127.0.0.1	global	ALL PRIVILEGES	Yes	Edit privileges Export
☐	root	:::1	global	ALL PRIVILEGES	Yes	Edit privileges Export
☐	root	localhost	global	ALL PRIVILEGES	Yes	Edit privileges Export

☐ Check all With selected: [Export](#)

New

[Add user account](#)

Console

Type here to search

31°C Haze 14:58 08-11-2023

CHECKING for PRIVILAGES

(for user2 who doesn't have
insert, update and delete privilege)

- Login as user 2 and perform the following operations:
 1. Display any table. (Perform select query on a table)
 2. Insert a row in any of the table in your database.
 3. Update a row in any of the table in your database.
 4. Now delete the row of any of the table in your database.

Error Messages when Accessed with Improper Privileges

- INSERT command denied to user 'user2'@localhost for table '***'
- UPDATE command denied to user 'user2'@localhost for table '***'
- DELETE command denied to user 'user2'@localhost for table '***'